Week 1 — Aug 19 & 21

Major Topics: Counting, sample spaces, permutations, combinations, multinomial counting

Aug 19

- Introductions, syllabus
- New York LOTTO example
- Multiplication rule
- Permutations and combinations

Aug 21

- First group quiz assigned (counting techniques)
- Homework discussion
- Review unclear homework problems
- Introduction to **multinomial counting**
- Group practice problems

Week 2 — Aug 26 & 28

Major Topics: Probability basics, sample spaces, algebra of events

Aug 26

- Homework 1 comments
- Challenge question: 3 heads in 8 flips vs 2 sixes in 5 rolls
- Probability via counting equally likely outcomes
- Sample space Ω, events
- Probability of an event
- Group quiz (finish on 8/28)

Aug 28

- Finish Quiz 1
- Homework comments: clear multiplication-rule structure
- Complete dice and coin activity
- **Algebra of events**: union, intersection, complement
- **Rules of probability**

- Note: infinite-outcome rule will come later

Week 3 — Sep 2 & 4

Major Topics: Independence, random variables, distributions, expectation

Sep 2

- Quiz 2
- Independence (informal → formal definition)
- Examples of independent / not independent events
- What is a random variable?
- Random variables as functions

Sep 4

- Quiz feedback
- Homework comments: describe sample spaces, not just counts
- Random variables
- **Distribution, Expected value, Variance, Standard deviation**
- Prep for Quiz 3

Week 4 — Sep 9 & 11

Major Topics: Expected value tools, LOTUS, variance, binomial distribution

Sep 9

- Quiz 3
- Expected value via distribution + linearity
- Variance and standard deviation
- **Law of the Unconscious Statistician (LOTUS)**

Sep 11

- Quiz 4 announcement
- Homework 7
- **Binomial distribution** (story, notation X~Bin(n,p))
- EV & Var formulas

- Why P(A∪B∪C) ≠ sum of probabilities (non-disjoint events)
- LOTUS again

Week 5 — Sep 16 & 18

Major Topics: Poisson distribution & approximation, geometric distribution

Sep 16

- Quiz
- Poisson as approximation to binomial (large n, small p)
- λ = np
- Poisson EV and Var = λ
- Poisson as a *model* (not equal-likelihood sample space)

Sep 18

- Return Quiz 4
- Homework 9
- Visualizing distributions
- More Poisson
- **Geometric(p)** distribution, story, EV, Var

Week 6 — Sep 23 & 25

Major Topics: Inclusion–exclusion, Quiz 5, Midterm 1

Sep 23

- Quiz 5
- Midterm prep
- Homework 10
- **Inclusion–Exclusion formula** (not on Midterm 1)

Sep 25

- **Midterm 1**

Week 7 — Sep 30 & Oct 2

Major Topics: Conditional probability, two-stage experiments

Sep 30

- Return Midterm 1
- Conditional probability definition + examples
- Card-drawing ace problem (urn interpretation)

Oct 2

- Quiz 6
- Homework 11
- Two-stage experiments
- Law of total probability

Week 8 — Oct 7 & 9

Major Topics: Bayes' formula, joint distributions, independence of RVs

Oct 7

- Bayes's formula
- Prior → likelihood → posterior
- Practice problems

Oct 9

- Quiz 7
- Joint distribution tables
- Marginals
- Independence of RVs
- Conditional probabilities from joint distributions

Week 9 — Oct 16

Major Topics: Continuous random variables, density functions

Oct 16

- Continuous RVs
- Density functions f(x)
- Expected value and variance via integrals
- Quiz on 10/21

Week 10 — Oct 21 & 23

Major Topics: Distribution functions, continuous distribution examples

Oct 21

- Quiz 8
- Distribution functions (CDFs)
- Medians of RVs

Oct 23

- Examples of continuous RVs
- Mole Day
- Prep for Midterm 2

Week 11 — Oct 28 & 30

Major Topics: Continuous practice, Midterm 2

Oct 28

- Quiz 9
- Practice with continuous distributions
- Breaking sticks, rainforest problems

Oct 30

- **Midterm 2**

Week 12 — Nov 4 & 6

Major Topics: Sums of RVs, covariance, Bernoulli indicators, variance formulas

Nov 4

- Indicators & Bernoulli RVs
- Sum of RVs
- E[X+Y]
- Write X as sum of indicator variables

Nov 6

- Quiz 10
- Covariance
- Variance of sum of RVs
- Apply to binomial

Week 13 — Nov 11 & 13

Major Topics: Law of Large Numbers, Chebyshev, CLT intro

Nov 11

- LLN intuition + Chebyshev's inequality
- Sequences of iid RVs
- Hypergeometric, Poisson, exponential, normal

Nov 13

- Quiz 11
- LLN continued
- **Central Limit Theorem (rough form)**
- Normal density; Z-table

Week 14 — Nov 18 & 20

Major Topics: Central Limit Theorem (formal), sums of standard distributions

Nov 18

- Quiz 12
- CLT precise statement
- Histogram correction examples

Nov 20

- Sums of:
    - Binomial
    - Poisson
    - Normal
    - Exponential (gamma)
    - Cauchy
- Quiz 12

Week 15 — Nov 25

Major Topics: Intro to statistics: p-values

Nov 25

- Sum of exponentials → gamma
- Applications to data
- **p-values**
- Confidence intervals (preview)

Week 16 — Dec 2

Major Topics: Confidence intervals, final exam prep

Dec 2

- Teaching evaluations
- Senior presentations
- Confidence intervals

- Final exam reminders